FusePlan — Presentation script

Read in full this is about 11 minutes; the highlighted parts plus a few connecting words make about 6, and the times next to each slide refer to that version (≈6–7 min in total). Everything else is there to pick from if time allows or for questions. Numbers come from the paper (Tables 1–4, Figures 4–6).

Slide 1 — Title (≈15 s)

We are Giuseppe Angelillis, Enrico Di Maria and Gabriele Germini. Our work is FusePlan, and the subtitle already says what it is about: when we fuse two models and the result improves, how much of that improvement is real information, and how much is just the effect of having two models instead of one.

Slide 2 — Problem and related work (≈50 s)

An architect rarely starts from a complete floor plan: they sketch a few rooms or reuse a fragment, then look for complete plans that contain it. Existing floor-plan retrieval methods (LayoutGMN, LayoutGKN, SSIG, Unit Region Encoding) learn a similarity between complete plans and are evaluated on complete plans. Partial plans do appear in the literature, in Graph2Plan and PlanCraft, but only for generation. So nobody retrieves floor plans from a partial query: that is our task. Concretely: we take a plan, remove some of its rooms, and ask the system to find the original among all 67,405 RPLAN plans.

A plan exists in two forms, an image and a graph of rooms, and the two fail in different ways: the image keeps the shape of the hole a removed room leaves, while a partial graph looks like the complete graph of some smaller plan. So fusing them seems natural. But any fusion of two models with partly independent errors improves, even if the second model knows nothing new. Multimodal methods like CrossOver and MMFE report only the fused accuracy, so they cannot tell the two cases apart. Our question is exactly this: does the image add information to the graph, or does it only act as a second model? From sketch retrieval we borrow the metric, an area under the reciprocal-rank curve over levels of completeness.

Slide 3 — Method (≈70 s)

FusePlan has two branches that search the same gallery and receive the same removed rooms, so we can compare them query by query.

The vision branch renders the plan as an image and encodes it with a frozen encoder, PE-Spatial, followed by GeM pooling and PCA whitening estimated on the training split, giving a 768-d vector. Optionally, we add a small head trained on the damage itself.

The graph branch turns each room into a node, with its type and its box, and each adjacency into an edge. A GATv2 encoder is trained with InfoNCE on pairs made of a damaged plan and its complete version. The pairs are asymmetric on purpose, because that is the task: map an incomplete plan onto the complete one. We also select the checkpoint on retrieval robustness, not on the loss, because the validation loss turned out to rank configurations in the opposite order of retrieval.

The fusion concatenates the two unit vectors weighted by the square roots of alpha and 1 − alpha: the inner product becomes a weighted sum of the two cosine similarities, so one exact FAISS search ranks the whole gallery. Alpha is chosen on validation.

The key idea is the ensemble control: we also fuse our graph with a second graph model, either a second training of the same network or a GraphSAGE encoder. That second model holds the same information as the graph, so whatever it gains is a pure ensemble effect. The complementarity gain, CG, is the fusion gain minus the gain of the stronger control: if it is positive, the image really brings something the graph does not have.

Slide 4 — Experiments (≈70 s)

A query is a plan with 25, 50 or 75 % of its rooms removed: in the image we fill those rooms with the background and erase their walls, in the graph we keep only the remaining rooms. For each query we take the reciprocal rank of the source plan and average it over the three levels: 1 means the source always comes first.

On the protocol: every choice, from configurations to checkpoints and fusion weights, is made on validation. Before reading the test split we wrote down our predictions, then read it once. This matters because choosing anything on the test set would inflate exactly the numbers we report.

Results. A room-type histogram finds almost nothing, so the task cannot be solved by just counting rooms. The frozen vision branch reaches 39.6, with the damage-trained head 53.3. LayoutGKN, retrained on the same pairs, reaches 61.1, our graph 90.3: 29 points more. The difference is where it happens: at 25 % removal the two are close, at 75 % LayoutGKN drops to 0.22 while ours still holds 0.79. A graph trained and selected for incompleteness is what makes the difference.

Every fusion improves on the graph, up to 94.1, and almost all of the gain comes at 75 % removal, the only level where the graph leaves room to improve. But a gain of +1.3 alone does not tell us whether the image helps or whether any second model would: this is what the next two slides answer.

Slide 5 — Relative strength and ablation (≈70 s)

In the plot, each point is one of the 37 graphs from our configuration search, fused with the vision branch: on the x axis the strength of the graph alone, on the y axis the fusion gain. The gain follows an inverted U. When the graph is much weaker than the image, the image dominates and

the graph adds little; when the two branches are equally strong the gain peaks, at about +0.20; when the graph approaches the ceiling there is little left to fix, and the gain falls to +0.013. So the fusion gain is not a property of the image: it depends on the relative strength of the two branches. This also explains our own history: an earlier version of the pipeline, with a weaker graph, measured +0.17; that point, the star, lies on the same curve, so the large old gain and the small new one do not contradict each other.

The ablations explain how we got there. On the graph, the temperature 0.1 is the strongest single change, from 52 to 75, but combining the useful changes brings it to 91, and above all at heavy damage: 0.81 against 0.38 at 75 % removal.

On the vision side, whitening helps every encoder, in all 26 settings. The cause: on synthetic renderings the frozen descriptors collapse into a narrow cone, so a damaged query looks similar to almost any plan; whitening spreads them out. And newer or larger encoders do not win: TIPSv2, the best encoder on complete plans in our benchmark, ranks sixth under damage.

Overall the search covered 37 graph configurations over four seeds and 52 vision configurations over eight encoders.

If there is time, a negative result: our first room-removal rendering erased the dark walls but kept the lighter interior ones, leaving each removed room as an empty outlined cell, which inflated a frozen encoder from 0.30 to 0.54. We caught it because we had pre-registered a check on other damages, and that is why vision configurations are now selected on the average of three damages.

Slide 6 — Complementarity gain (≈80 s)

This table is the answer to our question; numbers are averages over four replicas on the test split. A second copy of our graph, fused with the graph, already gains +1.63 points.

The frozen vision branch gains +1.25, less than the control: CG −0.38. So, with a strong graph, a frozen image encoder adds nothing beyond an ensemble effect, while the raw +1.25 alone would have suggested otherwise.

The two heads trained on the damage gain +2.79 and +3.66, and exceed the control by +1.16 and +2.03, positive in all four replicas: they carry information of their own. But even here the control explains about half of the gain, so reporting the raw gain overstates the contribution of the image by roughly a factor of two.

Where does the image help? Splitting the queries, the CG is zero at 25 and 50 % removal and becomes positive only at 75 %, and it is largest on small plans, with five rooms or fewer, where even the frozen branch passes the control. The cause is the one we started from: when few rooms remain, the partial graph is compatible with many plans, while the image still keeps the shape of the hole. This part is a post-hoc analysis on one seed, so we read it as a description. Consistently,

with graphs only as strong as the image, even the frozen vision branch passes the controls by 8 to 16 points.

Slide 7 — Conclusions (≈50 s)

Three conclusions. First, a graph trained on partial plans retrieves a plan from half of its rooms almost as well as from a quarter missing (0.954 against 0.963).

Second, the image adds information of its own only under two conditions: it is trained on the damage, or the graph is no stronger than the image.

Third, a recommendation that goes beyond floor plans: part of any fusion gain comes just from combining two models, so a fusion gain should always be compared with an ensemble control, here a fusion of two graph models, to measure what the second modality really adds.

Limitations: we use only RPLAN, with synthetic damages, and our graphs come from exact records, not from parsed images. Future work: graphs parsed from images, external datasets, a learned or per-query fusion, and human relevance judgements. Thank you.